

Task 01 Calculations

Group: B-1

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Questions

1. Using $f_{TIM6_CLK} = 90MHz$, determine PSC and ARR values that produce the following TIM6 overflow periods. For each target, show at least two valid (PSC, ARR) pairs and state which you would choose and why.

Target Period	PSC	ARR	Actual Period
1ms			
100 μ s			
500ms			
1s			

2. For TIM3 generating 1kHz PWM on APB1 ($f_{TIM3} = 90MHz$):
- Determine PSC and ARR.
 - Compute the CCR1 value required for duty cycles of 25%, 50%, 75%, and 100%.
 - Compute the CCR1 value for an arbitrary duty cycle D in terms of ARR.
3. For TIM1 driving a WS2812B LED at $f_{TIM1} = 180MHz$, PSC = 0:
- Calculate ARR so that one-bit period equals 1250ns.
 - Calculate CCR1 for a logic-1 bit (T1H $\approx$ 800ns).
 - Calculate CCR1 for a logic-0 bit (T0H $\approx$ 400ns).
 - Verify that your T1H and T0L values are within the WS2812B tolerance given in table.

Symbol	Nominal	Tolerance	Meaning
T0H	400ns	$\pm 150ns$	High time for a logic 0 bit
T0L	850ns	$\pm 150ns$	Low time for a logic 0 bit
T1H	800ns	$\pm 150ns$	High time for a logic 1 bit
T1L	450ns	$\pm 150ns$	Low time for a logic 1 bit
RES	>50 μ s	---	Reset latch: data line held LOW after last bit

4. The DWT cycle counter runs at $f_{CPU} = 180MHz$. Convert the following cycle counts to ns, μ s, and ms: 3600, 90000, 9000000.

Solutions

1.

a) Given,

$$f_{TIM6_CLK} = 90MHz$$

$$T_{Period} = 1ms$$

We know,

$$T_{Period} = \frac{(ARR + 1)(PSC + 1)}{f_{TIM6_CLK}}$$

$$Or, (ARR + 1)(PSC + 1) = T_{Period} \times f_{TIM6_CLK}$$

$$Or, (ARR + 1)(PSC + 1) = 0.001 \times 90000000$$

$$Or, (ARR + 1)(PSC + 1) = 90000$$

Set 1: ARR = 8999 and PSC = 9 so that ARR+1 times PSC+1 = 90000

Set 2: ARR = 44999 and PSC = 1, this pair also works

The second set is better because, it initially scales the frequency to 45MHz, which gives the tick-size almost 0.022μs so we can count lower periods later on.

b) Given,

$$f_{TIM6_CLK} = 90MHz$$

$$T_{Period} = 100\mu s$$

We know,

$$T_{Period} = \frac{(ARR + 1)(PSC + 1)}{f_{TIM6_CLK}}$$

$$Or, (ARR + 1)(PSC + 1) = T_{Period} \times f_{TIM6_CLK}$$

$$Or, (ARR + 1)(PSC + 1) = 100 \times 10^{-6} \times 90000000$$

$$Or, (ARR + 1)(PSC + 1) = 9000$$

Set 1: ARR = 4499 and PSC = 1

Set 2: ARR = 999 and PSC = 8

The first pair is suitable as it gives tick at almost 0.022μs time, so the chunk is short and accurate for flexible counting.

c) Given,

$$f_{TIM6_CLK} = 90MHz$$

$$T_{Period} = 500ms$$

We know,

$$T_{Period} = \frac{(ARR + 1)(PSC + 1)}{f_{TIM6_CLK}}$$

$$Or, (ARR + 1)(PSC + 1) = T_{Period} \times f_{TIM6_CLK}$$

$$Or, (ARR + 1)(PSC + 1) = 0.5 \times 90000000$$

$$Or, (ARR + 1)(PSC + 1) = 45 \times 10^6$$

$$\text{Set 1: } ARR = 59999, PSC = 749$$

$$\text{Set 2: } ARR = 65502, PSC = 687$$

In this case also, Set 1 is better for fine-tuning later on and gives the exact timing. But in Set 2, it has better resolution ($8.33\mu s$ vs $7.64\mu s$) but doesn't hit the target time (500.4ms almost). So, both have tradeoffs but I would choose Set 1 for this case as there are very little benefits from Set 2.

d) Given,

$$f_{TIM6_CLK} = 90MHz$$

$$T_{Period} = 1s$$

We know,

$$T_{Period} = \frac{(ARR + 1)(PSC + 1)}{f_{TIM6_CLK}}$$

$$Or, (ARR + 1)(PSC + 1) = T_{Period} \times f_{TIM6_CLK}$$

$$Or, (ARR + 1)(PSC + 1) = 1 \times 90000000$$

$$Or, (ARR + 1)(PSC + 1) = 90 \times 10^6$$

$$\text{Set 1: } ARR = 65529 \text{ and } PSC = 1373$$

$$\text{Set 2: } ARR = 59999 \text{ and } PSC = 1499$$

The second pair here is better because of the precision of the Target Period. First pair gives the time slightly more, but their resolution difference is about $1.4\mu s$, so it is better to choose Set 1 for precise operation.

Finally, all pairs used here create the same target period mentioned in the table:

Target Period	PSC	ARR	Actual Period
1ms	1	44999	1ms
100 μs	1	4499	100 μs
500ms	749	59999	500ms
1s	1499	59999	1s

2. a) Given,

$$f_{TIM3} = 90MHz$$

$$f_{PWM} = 1kHz$$

We know,

$$f_{PWM} = \frac{f_{TIM3}}{(ARR + 1)(PSC + 1)}$$

$$Or, (ARR + 1)(PSC + 1) = \frac{f_{TIM3}}{f_{PWM}}$$

$$Or, (ARR + 1)(PSC + 1) = \frac{90000000}{1000}$$

$$Or, (ARR + 1)(PSC + 1) = 90000$$

For this PWM output, $ARR = 44999$ and $PSC = 1$ is a good choice

b) From (a), $ARR = 44999$

$$\therefore \text{For Duty Cycle} = 25\%, CCR1 = \frac{D}{100} \times (ARR + 1) = 11250$$

$$\text{For Duty Cycle} = 50\%, CCR1 = \frac{D}{100} \times (ARR + 1) = 22500$$

$$\text{For Duty Cycle} = 75\%, CCR1 = \frac{D}{100} \times (ARR + 1) = 33750$$

$$\text{For Duty Cycle} = 100\%, CCR1 = \frac{D}{100} \times (ARR + 1) = 45000$$

c) For any arbitrary CCR1 value,

$$CCR1 = \frac{D}{100} \times (ARR + 1), \text{ where } D = \text{Duty Cycle in Percentage}$$

3. a) Given,

$$f_{TIM1_CLK} = 180MHz$$

$$\therefore T_{TIM1_CLK} = \frac{1}{180000000} \approx 5.56ns$$

Now,

$$\text{Number of Ticks} \approx \frac{1250}{5.56} \approx 225 \text{ and } ARR \approx 225 - 1 \approx 224 \text{ (Ans.)}$$

b) Given,

$$T_{T1H} \approx 800ns$$

$$T_{Period} = 1250ns$$

$$\text{From (a), } ARR \approx 224$$

$$\therefore CCR1 = \frac{T_{T1H}}{T_{Period}} \times (ARR + 1) \approx \frac{800}{1250} \times 225 \approx 144 \text{ (Ans.)}$$

c) Given,

$$T_{T0H} \approx 400ns$$

$$T_{Period} = 1250ns$$

$$\text{From (a), } ARR \approx 224$$

$$\therefore CCR1 = \frac{T_{T0H}}{T_{Period}} \times (ARR + 1) \approx \frac{400}{1250} \times 225 \approx 72 \text{ (Ans.)}$$

$$\text{d) From (a), } T_{TIM1_CLK} \approx 5.56ns$$

$$\text{For } CCR1 \approx 144,$$

$$T1H \approx (144 \times 5.56)ns \approx 800.64ns \approx 800ns$$

Again,

$$\text{For } CCR1 \approx 72,$$

$$T0H \approx (72 \times 5.56)ns \approx 400.32ns \approx 400ns$$

This verifies that the CCR1 values lie between the threshold of T0H and T1H i.e. 250-550ns and 650-950ns. For the LOW values,

$$T1L \approx (1250 - 800)ns \approx 450ns [\text{Between the } 300 - 600ns \text{ threshold}]$$

$$T0L \approx (1250 - 400)ns \approx 850ns [\text{Between the } 600 - 900ns \text{ threshold}]$$

Finally, all values cover the tolerance threshold mentioned in the table.

4. Given,

$$f_{CPU} = 180MHz$$

a) For Cycle = 3600,

$$t_{ns} = \frac{Cycle \times 1000}{f_{CPU(MHz)}} = \frac{3600 \times 1000}{180} = 20000ns$$

$$t_{\mu s} = \frac{Cycle}{f_{CPU(MHz)}} = \frac{3600}{180} = 20\mu s$$

$$t_{ms} = \frac{Cycle}{f_{CPU(MHz)} \times 1000} = \frac{3600}{180 \times 1000} = 0.02ms (Ans.)$$

b) For Cycle = 90000,

$$t_{ns} = \frac{Cycle \times 1000}{f_{CPU(MHz)}} = \frac{90000 \times 1000}{180} = 500000ns$$

$$t_{\mu s} = \frac{Cycle}{f_{CPU(MHz)}} = \frac{90000}{180} = 500\mu s$$

$$t_{ms} = \frac{Cycle}{f_{CPU(MHz)} \times 1000} = \frac{90000}{180 \times 1000} = 0.5ms (Ans.)$$

c) For Cycle = 9000000,

$$t_{ns} = \frac{Cycle \times 1000}{f_{CPU(MHz)}} = \frac{9000000 \times 1000}{180} = 50000000ns$$

$$t_{\mu s} = \frac{Cycle}{f_{CPU(MHz)}} = \frac{9000000}{180} = 50000\mu s$$

$$t_{ms} = \frac{Cycle}{f_{CPU(MHz)} \times 1000} = \frac{9000000}{180 \times 1000} = 500ms (Ans.)$$

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